



Big data frameworks demo requirements

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Outline

- Apache Hadoop (HDFS, MapReduce, YARN)
- Apache Spark
- Scalable Distributed Data Anonymization



Deployment environments

- Docker Compose
- Kubernetes

NOTE: Considering the limited network bandwidth available during class, please download the necessary Docker images before the lecture



Docker Compose requirements

Just install it! duh



<https://docs.docker.com/compose/install/>

Kubernetes requirements

Docker

Platform for developing, shipping, and running applications



<https://docs.docker.com/get-docker/>

kind

Tool for running local Kubernetes clusters



<https://kind.sigs.k8s.io/docs/user/quick-start/>

kubectl

Kubernetes command-line tool



<https://kubernetes.io/docs/tasks/tools/>



Helm

Package manager for Kubernetes



<https://helm.sh/docs/intro/install/>

Apache Hadoop

Docker compose deployment



[https://github.com/big-data-europe/
docker-hadoop](https://github.com/big-data-europe/docker-hadoop)

Kubernetes deployment



[https://github.com/matthewrossi/
adm-laboratory-hadoop](https://github.com/matthewrossi/adm-laboratory-hadoop)

Apache Spark

Docker compose deployment



[https://github.com/big-data-europe/
docker-spark](https://github.com/big-data-europe/docker-spark)

Kubernetes deployment



[https://github.com/matthewrossi/
adm-laboratory-spark](https://github.com/matthewrossi/adm-laboratory-spark)

Scalable Distributed Data Anonymization

Follow the instructions in the repository and make sure to checkout the endpoint of the services



<https://github.com/mosaicrown/mondrian>